

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True)

```
import pandas as pd

file_path = "/content/dataset1.csv"
data1= pd.read_csv(file_path)
data1.head()
```

```
data1.head()
```

```
data1.isnull().sum()
```

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_2004/1737738369.py in <cell line: 0>()
----> 1 data1.isnull().sum()

NameError: name 'data1' is not defined
```

```
print(data1.duplicated().sum())
```

```
print(data1.describe())
```

```
data1.columns = data1.columns.str.strip()
```

```
print(data1.isnull().sum())
print(data1.duplicated().sum())
print(data1.info())
```

```
import matplotlib.pyplot as plt

plt.plot(df['Maths'], label='Maths')
plt.plot(df['Science'], label='Science')
plt.plot(df['English'], label='English')

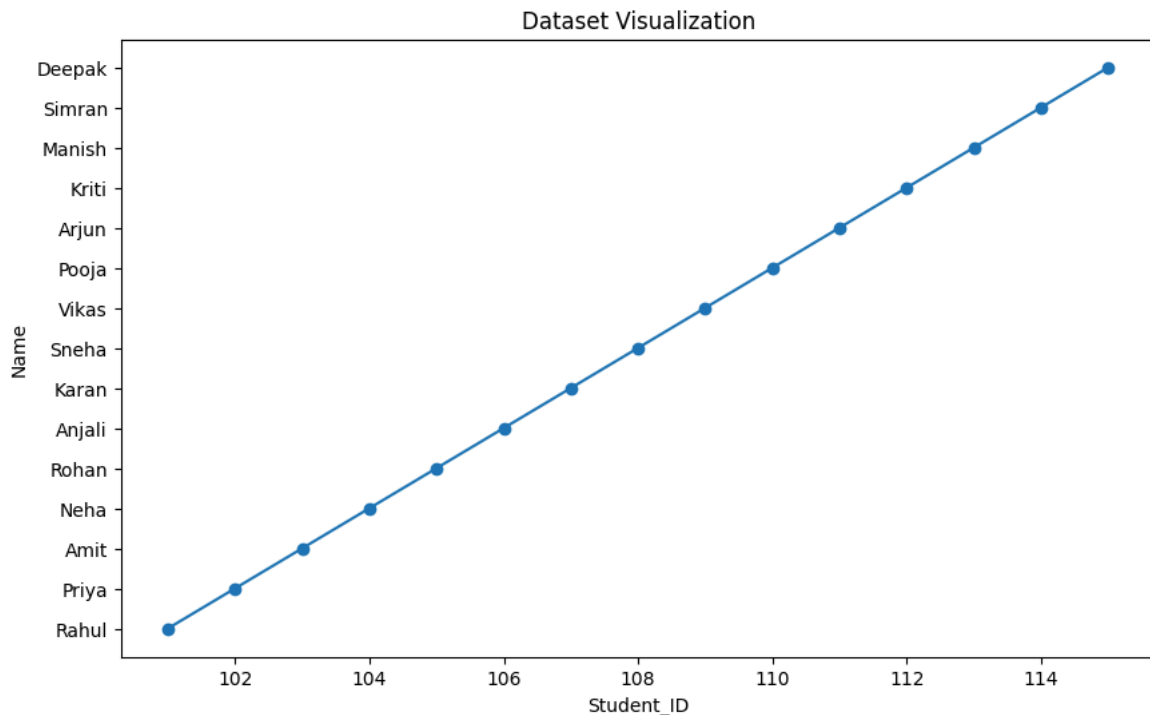
plt.title("Subject-wise Marks Comparison")
plt.xlabel("Student Index")
plt.ylabel("Marks")
plt.legend()
plt.grid(True)

plt.show()
```

	Student_ID	Name	Gender	Age	City	Department	Semester	Maths	\
0	101	Rahul	Male	20	Delhi	Btech	2	85.0	
1	102	Priya	Female	21	Mumbai	BCA	2	76.0	
2	103	Amit	Male	22	Delhi	Btech	4	92.0	
3	104	Neha	Female	20	Pune	BBA	2	65.0	
4	105	Rohan	Male	23	Delhi	MCA	4	81.0	

	Science	English	Attendance	Fees	Placement
0	78.0	90.0	92.0	50000.0	Yes
1	88.0	84.0	95.0	52000.0	No
2	91.0	89.0	88.0	60000.0	Yes
3	72.0	80.0	90.0	45000.0	No
4	85.0	NaN	87.0	61000.0	Yes

```
Index(['Student_ID', 'Name', 'Gender', 'Age', 'City', 'Department', 'Semester',
      'Maths', 'Science', 'English', 'Attendance', 'Fees', 'Placement'],
      dtype='object')
```



```
import matplotlib.pyplot as plt

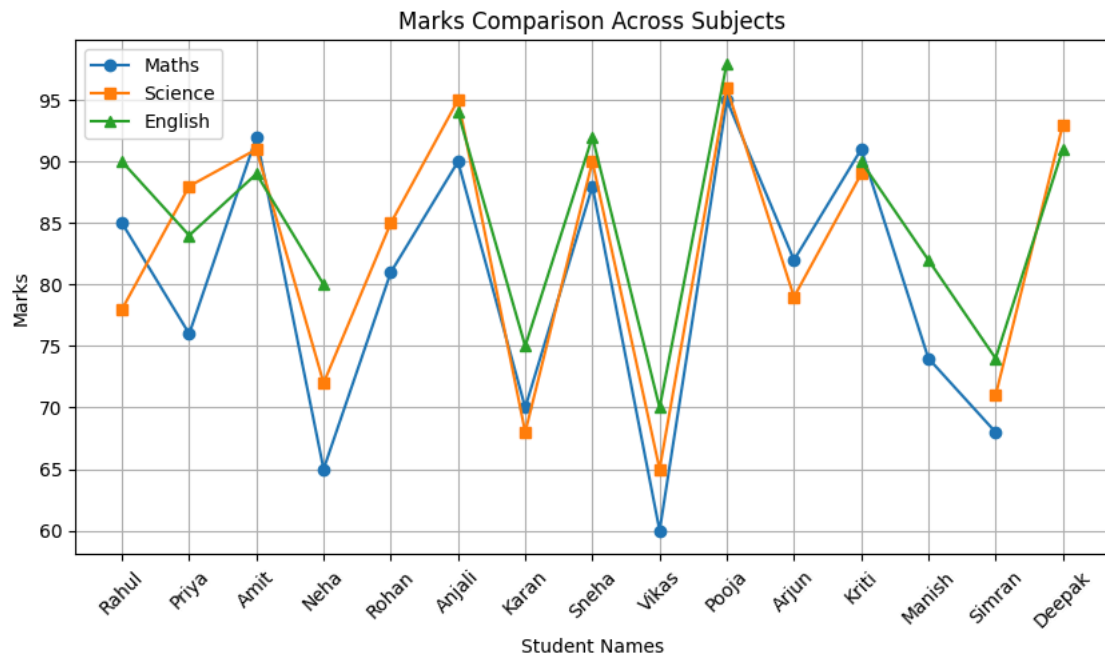
plt.figure(figsize=(10,5))

plt.plot(df['Name'], df['Maths'], marker='o', label='Maths')
plt.plot(df['Name'], df['Science'], marker='s', label='Science')
plt.plot(df['Name'], df['English'], marker='^', label='English')

plt.title("Marks Comparison Across Subjects")
plt.xlabel("Student Names")
plt.ylabel("Marks")
plt.legend()

plt.grid(True)

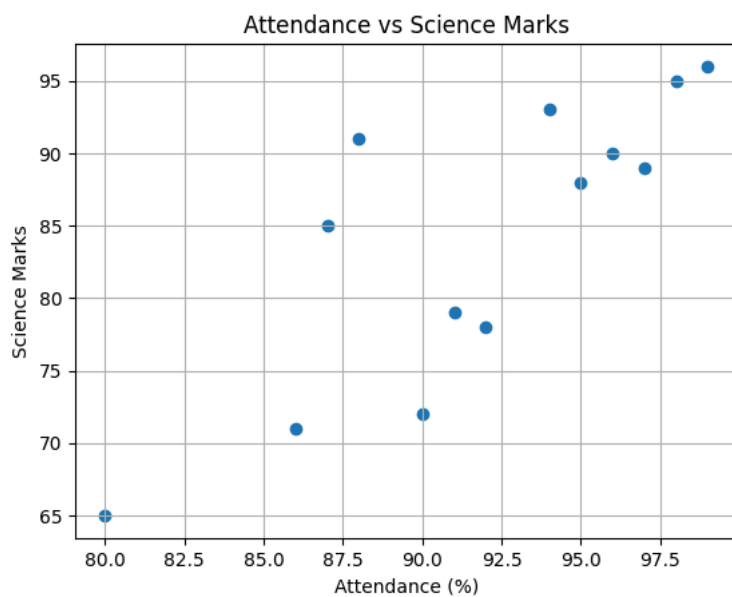
plt.xticks(rotation=45)
plt.show()
```



```
plt.scatter(df['Attendance'], df['Science'])

plt.title("Attendance vs Science Marks")
plt.xlabel("Attendance (%)")
plt.ylabel("Science Marks")
plt.grid(True)

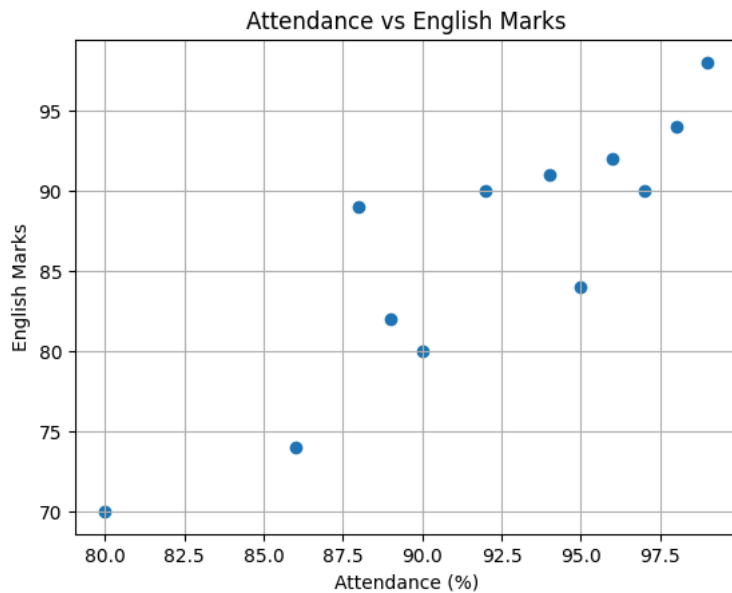
plt.show()
```



```
plt.scatter(df['Attendance'], df['English'])

plt.title("Attendance vs English Marks")
plt.xlabel("Attendance (%)")
plt.ylabel("English Marks")
plt.grid(True)

plt.show()
```



```
import matplotlib.pyplot as plt
import numpy as np

# Missing values hatao
temp_df = df[['Attendance', 'Maths']].dropna()

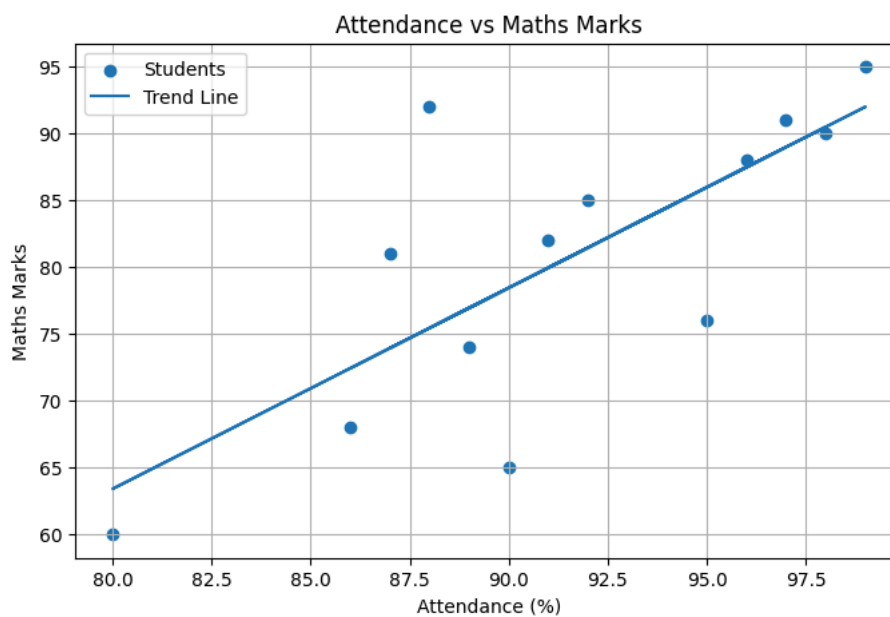
# Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(temp_df['Attendance'], temp_df['Maths'], label='Students')

# Trend Line
z = np.polyfit(temp_df['Attendance'], temp_df['Maths'], 1)
p = np.poly1d(z)

plt.plot(
    temp_df['Attendance'],
    p(temp_df['Attendance']),
    label='Trend Line'
)

plt.title("Attendance vs Maths Marks")
plt.xlabel("Attendance (%)")
plt.ylabel("Maths Marks")
plt.legend()
plt.grid(True)

plt.show()
```

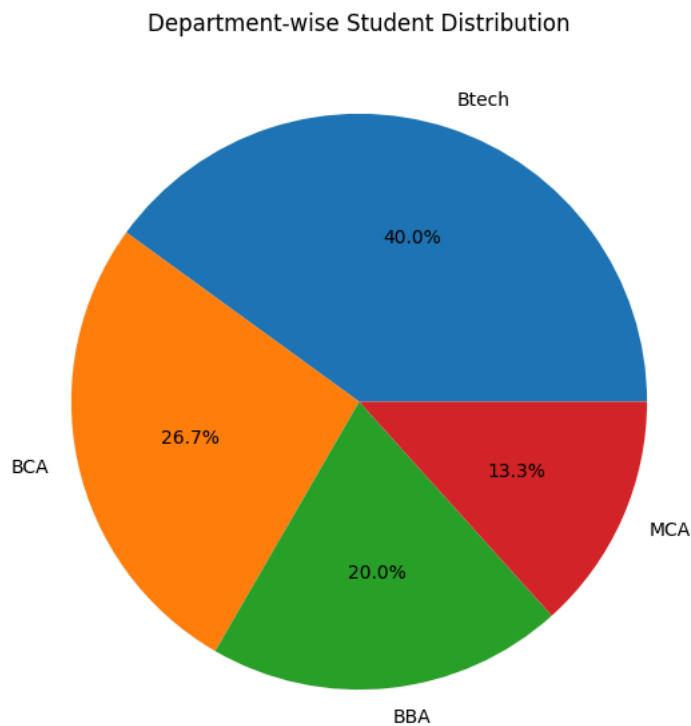


```
import matplotlib.pyplot as plt
```

```
dept_count = df['Department'].value_counts()

plt.figure(figsize=(7,7))
plt.pie(
    dept_count,
    labels=dept_count.index,
    autopct='%1.1f%%'
)

plt.title("Department-wise Student Distribution")
plt.show()
```



```
import matplotlib.pyplot as plt

dept_count = df['Department'].value_counts()

plt.figure(figsize=(8,5))
plt.bar(dept_count.index, dept_count.values)

plt.title("Students by Department")
plt.xlabel("Department")
plt.ylabel("Number of Students")
plt.grid(True)

plt.show()
```

Students by Department

```
import matplotlib.pyplot as plt  
  
plt.figure(figsize=(8,5))
```